HDG method for P-Laplacian Problem

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1 Problem

In this paper we consider the HDG method for P-Laplacian problem

$$\begin{aligned} -\nabla\cdot(\,|\nabla u|^{p-2}\,\nabla u\,) &= f \quad \text{in } \Omega,\\ u &= g \quad \text{on } \partial\Omega, \end{aligned}$$

Using the first variation it is wasy to see the solution of the P-Laplacian equation is the minimizer of the following minimizing problem

$$J(u)=\min_{A} J(v), \tag{1.1}$$

where the energy functional

$$J(v)=\frac{1}{p}\int_\Omega |\nabla v|^p dx-\int_\Omega fv dx \tag{1.2}$$

and the admissible set $A=\{w\in W^{1,p}(\Omega)\,\text{and}\,w=g\,\text{on}\,\partial\Omega\}$.

2 Prelimilaries

2.1 HDG method and minimization problem

Let us begin with the basic finite dimensional spaces associated with the shape regular triangulation $\mathcal{T}_h=\{K\}$ of the domain Ω, whose boundaries are labeled as F. The union of all faces (interior faces) will be denoted as Γ (Γ^o). And Γ^d is the set of all boundary faces. The following approximation spaces of HDG method will be used

$$\begin{aligned} \boldsymbol{V}_h &:= \{\boldsymbol{v}\in \boldsymbol{L}^2(\mathcal{T}_h): \quad \boldsymbol{v}|_K\in\boldsymbol{\mathcal{P}}_k(K) \quad \forall\, K\in\mathcal{T}_h\},\\ W_h &:= \{w\in L^2(\mathcal{T}_h): \quad w|_K\in\mathcal{P}_k(K) \quad \forall\, K\in\mathcal{T}_h\},\\ M_h &:= \{\mu\in L^2(\mathcal{E}_h): \quad \mu|_F\in\mathcal{P}_k(F) \quad \forall\, F\in\Gamma\},\\ M_h^g &:= \{\mu\in M_h: \quad \mu|_F=g_h \quad \forall\, F\in\Gamma^d\}, \end{aligned}$$

respectively, where g_h is the normal L^2-projection of data g on the boundary edges. $\boldsymbol{\mathcal{P}}_k(K):=[\mathcal{P}_k(K)]^n$, $\mathcal{P}_k(K)\,\text{and}\,\mathcal{P}_k(F)$ are polynomial functions of degree at most $k\geq 0$. As usual, the vector-valued space $\boldsymbol{\mathcal{P}}_k(K)$ yields a auxiliary vector approximation of $\boldsymbol{q}_h$ to the negative derivative, $-\nabla u$, while the scalar space W_h and M_h^g provide an approximation u_h to the exact solution u in the element, and a numerical flux $\widehat{u}_h$ to the trace of u on the element boundary.

In this subsequent paper, we still consider the following HDG method for p-laplacian equation: Find $(\boldsymbol{q}_h,\boldsymbol{\sigma}_h,u_h,\widehat{u}_h)\in\boldsymbol{V}_h\times\boldsymbol{V}_h\times W_h\times M_h$ such that

$$(\boldsymbol{q}_h,\boldsymbol{r})_{\mathcal{T}_h}-(u_h,\nabla\cdot\boldsymbol{r})_{\mathcal{T}_h}+\langle\widehat{u}_h,\boldsymbol{r}\cdot\boldsymbol{n}\rangle_{\partial\mathcal{T}_h}= \quad 0, \tag{2.1a}$$

$$(\boldsymbol{\sigma}_h,\boldsymbol{r})_{\mathcal{T}_h}-(|\boldsymbol{q}_h|^{p-2}\,\boldsymbol{q}_h,\boldsymbol{r})_{\mathcal{T}_h}= \quad 0, \tag{2.1b}$$

$$-(\boldsymbol{\sigma}_h, \nabla w)_{\mathcal{T}_h} + \langle \widehat{\boldsymbol{\sigma}}_h \cdot \boldsymbol{n}, w\rangle_{\partial\mathcal{T}_h} - (f, w)_{\mathcal{T}_h} = \quad 0, \tag{2.1c}$$

$$\langle \widehat{u}_h, \mu\rangle_{\partial\Omega} - \langle g, \mu\rangle_{\partial\Omega} = \quad 0, \tag{2.1d}$$

$$\langle \widehat{\boldsymbol{\sigma}}_h \cdot \boldsymbol{n}, \mu\rangle_{\partial\mathcal{T}_h\backslash\partial\Omega} = \quad 0, \tag{2.1e}$$

for all $(\boldsymbol{r}, w, \mu) \in \boldsymbol{V}_h \times W_h \times M_h$, where $\widehat{\boldsymbol{\sigma}}_h \cdot \boldsymbol{n} = \boldsymbol{\sigma}_h \cdot \boldsymbol{n} + \tau(u_h - \widehat{u}_h)$. The stablization parameter τ is piecewise constant. Notations of integrals above in the weak form and throughout the paper will be written in short as

$$(u, w)_{\mathcal{T}_h} = \sum_{K\in\mathcal{T}_h} (u, w)_K \quad \text{and} \quad (u, w)_{\partial\mathcal{T}_h} = \sum_{K\in\partial\mathcal{T}_h} (u, w)_{\partial K}.$$

Without loss of generalization, for any p $\in [1, \infty)$ we will use standard Sobolev space $W^{1,p}(\Omega)$, $H^k(\Omega)$ and $L^p(\Omega)$ with their corresponding norms. Similarly, self-evident notations are applied for vector-valued functions.

The associated discrete functional proposed in [?] is in the form of

$$J_h(v_h, \hat{v}_h; \boldsymbol{q}_h) = \frac{1}{p}\int_\Omega |\boldsymbol{q}_h|^p \, dx + \frac{1}{2}\langle \tau(v_h - \widehat{v}_h), v_h - \widehat{v}_h\rangle_{\partial\mathcal{T}_h} - \int_\Omega f v_h \, dx, \tag{2.2}$$

where function $\boldsymbol{q}_h(v_h, \hat{v}_h)$ is determined by

$$(\boldsymbol{q}_h, \boldsymbol{r})_{\mathcal{T}_h} - (v_h, \nabla\cdot \boldsymbol{r})_{\mathcal{T}_h} + \langle \widehat{v}_h, \boldsymbol{r}\cdot\boldsymbol{n}\rangle_{\partial\mathcal{T}_h} = 0, , \quad \forall \boldsymbol{r} \in \boldsymbol{V}_h. \tag{2.3}$$

For simplicity we sometimes just write $\boldsymbol{q}_h$ for short. The discrete problem is to seek a minimizer $(u_h, \hat{u}_h)$ of (2.2) among all possible vectors $(v_h, \hat{v}_h)$ in the spaces $W_h \times M_h^g$, i, e

$$J_h(u_h, \hat{u}_h; \boldsymbol{q}_h) = \min_{W_h\times M_h^g} J_h(v_h, \hat{v}_h; \boldsymbol{q}_h) \tag{2.4}$$

We will verify in the Appendix [?] that the minimizer to the discrete problem is the solution to the HDG weak form. Equivalently this is to say that the HDG form of equation (2.1) is the Euler-Lagrange equation by using first variation. Therefore, if we can show the variational convergence of the discrete approximations to the minimizing problem we actually prove the convergence of the numerical approximations to the HDG form.

Add the reason why we consider the minimization problem (both from the computational and theoretical aspect).

2.2 Trace operator and primal formulation

Let us first introduce the conventional element boundary-valued trace operators, the jump, $[\![\cdot]\!]$, and the average, $\{\!\{\cdot\}\!\}$, to help us manipulate the boundary terms and obtain some primal formulations. The advantage of these operator is that we could split the boundary stability term into effect of the jumps and the averages, and together with two lifting operator defined in the next section we could explicitly express the negative weak derivative $\boldsymbol{q}_h$ in terms of u_h. This key step permits the possibility to use the classical Sobolev theory to analyze variational problem under the HDG framework. We only list the definitions below

$$\begin{aligned}
&\{\!\{w\}\!\} = \frac{w^+ + w^-}{2}, && \{\!\{\boldsymbol{r}\}\!\} = \frac{\boldsymbol{r}^+ + \boldsymbol{r}^-}{2}, && \forall e \in \Gamma^o,\\
&[\![\boldsymbol{r}]\!] = \boldsymbol{r}^+\cdot\boldsymbol{n}^+ + \boldsymbol{r}^-\cdot\boldsymbol{n}^-, && [\![w]\!] = w^+\boldsymbol{n}^+ + w^-\boldsymbol{n}^-, && \forall e \in \Gamma^o,\\
&[\![\boldsymbol{r}]\!] = \boldsymbol{r}\cdot\boldsymbol{n}, && [\![w]\!] = w\boldsymbol{n}, \ \{\!\{w\}\!\} = w, \ \{\!\{\boldsymbol{r}\}\!\} = \boldsymbol{r}, && \forall e \in \Gamma_d,
\end{aligned}$$

for all $(w, \boldsymbol{r}) \in W_h \otimes \boldsymbol{V}_h$. See [?] for more details.

Now a straightforward computation shows that the stabilization term $\langle \tau(v_h - \widehat{v}_h), v_h - \widehat{v}_h\rangle_{\partial\mathcal{T}_h}$ appeared in the discrete functional J_h can be rewritten in terms of the average and jump.

Lemma 2.1. *let* $(v_h, \hat{v}_h) \in W_h \times M_h^g$

$$\begin{aligned}
&\langle \tau(v_h - \widehat{v}_h), v_h - \widehat{v}_h \rangle_{\partial \mathcal{T}_h} \\
&= \langle \tau(v_h - g), v_h - g \rangle_{\Gamma_d} + \frac{1}{2} \langle \tau [\![v_h]\!], [\![v_h]\!] \rangle_{\Gamma^o} + 2 \langle \tau \{\!\{ v_h - \hat{v}_h \}\!\}, \{\!\{ v_h - \hat{v}_h \}\!\} \rangle_{\Gamma^o}. \qquad (2.5)
\end{aligned}$$

Proof. We notice that

$$\langle \tau(v_h - \widehat{v}_h), v_h - \widehat{v}_h \rangle_{\partial \mathcal{T}_h} = \sum_K \langle \tau(v_h - \widehat{v}_h), v_h - \widehat{v}_h \rangle_{\partial K}.$$

Then we separate the quantities above into the contributions from the boundary edges and the interior edges

$$\begin{aligned}
&= \sum_{e \in \Gamma_d} \langle \tau(v_h - \widehat{v}_h), v_h - \widehat{v}_h \rangle_e + \sum_{e \in \Gamma^o} \left(\langle \tau(v_h - \widehat{v}_h), v_h - \widehat{v}_h \rangle_{e^+} + \langle \tau(v_h - \widehat{v}_h), v_h - \widehat{v}_h \rangle_{e^-} \right), \\
&= (I) + (II).
\end{aligned}$$

By the restriction of the space M_h^g and g_h is normal L^2- projection, hence we have

$$\begin{aligned}
(I) &= \sum_{e \in \Gamma_d} \langle \tau(v_h - g_h), v_h - g_h \rangle_e \\
&= \sum_{e \in \Gamma_d} \tau \langle P(v_h - g), P(v_h - g) \rangle_e \\
&= \sum_{e \in \Gamma_d} \langle \tau(v_h - g), v_h - g \rangle_e \\
&= \langle \tau(v_h - g), v_h - g \rangle_{\Gamma_d}.
\end{aligned}$$

Suppose the interior edge e is shared by element K^+ and K^-, then,

$$(II) = \sum_{e \in \Gamma_o} \int_{e \in K^+ \bigcap K^-} \tau^+ (v_h^+ - \widehat{v}_h)(v_h^+ - \widehat{v}_h) + \tau^- (v_h^- - \widehat{v}_h)(v_h^- - \widehat{v}_h) ds.$$

Because the stabilization parameter τ considered in this paper is single valued piecewise constant, then $\tau^+ = \tau^- = \tau$ and

$$(II) = \sum_{e \in \Gamma_o} \tau \int_{e \in K^+ \bigcap K^-} (v_h^+ - \widehat{v}_h)(v_h^+ - \widehat{v}_h) + (v_h^- - \widehat{v}_h)(v_h^- - \widehat{v}_h) ds.$$

Further, we can simplify the integrand as

$$\begin{aligned}
&(v_h^+ - \widehat{v}_h)(v_h^+ - \widehat{v}_h) + (v_h^- - \widehat{v}_h)(v_h^- - \widehat{v}_h) \\
&= v_h^+ v_h^+ + v_h^- v_h^- - 2\widehat{v}_h (v_h^+ + v_h^-) + 2\widehat{v}_h^2 \\
&= \frac{1}{2} \left[(v_h^+ + v_h^-)^2 + (v_h^+ - v_h^-)^2 \right] - 4\widehat{v}_h \{\!\{ v_h \}\!\} + 2\widehat{v}_h^2 \\
&= \frac{1}{2} [\![v_h]\!] \cdot [\![v_h]\!] + 2 \{\!\{ v_h - \hat{v}_h \}\!\}^2.
\end{aligned}$$

Therefore,

$$\begin{aligned}
(II) &= \sum_{e \in \Gamma_o} \int_{e \in K^+ \bigcap K^-} \tau \left(\frac{1}{2} [\![v_h]\!] \cdot [\![v_h]\!] + 2 \{\!\{ v_h - \hat{v}_h \}\!\}^2 \right) ds \\
&= \frac{1}{2} \langle \tau [\![v_h]\!], [\![v_h]\!] \rangle_{\Gamma^o} + 2 \langle \tau \{\!\{ v_h - \hat{v}_h \}\!\}, \{\!\{ v_h - \hat{v}_h \}\!\} \rangle_{\Gamma^o}.
\end{aligned}$$

Combine the result for (I) and (II) we obtain (2.5). □

In order to explicitly express $\boldsymbol{q}_h(v_h, \hat{v}_h)$ defined in the weak form (2.3) it comes out that we have to deal with the boundary integration in the form of $\langle v_h - \widehat{v}_h, \boldsymbol{r}\cdot\boldsymbol{n}\rangle_{\partial\mathcal{T}_h}$.

Lemma 2.2. *For each* $(v_h, \hat{v}_h) \in W_h \times M_h$

$$\langle v_h - \widehat{v}_h, \boldsymbol{r}\cdot\boldsymbol{n}\rangle_{\partial\mathcal{T}_h} = \langle [\![v_h - \widehat{v}_h]\!], \{\!\{\boldsymbol{r}\}\!\}\rangle_{\Gamma} + \langle \{\!\{v_h - \widehat{v}_h\}\!\}, [\![\boldsymbol{r}]\!]\rangle_{\Gamma^o}, \quad \forall \boldsymbol{r} \in \boldsymbol{V}_h. \tag{2.6}$$

Proof. Similar as Lemma 2.1 we first break the term into the quantities from the boundary edges and interior edges

$$\begin{aligned}\langle v_h - \widehat{v}_h, \boldsymbol{r}\cdot\boldsymbol{n}\rangle_{\partial\mathcal{T}_h} &= \sum_{e\in\Gamma_d}\langle v_h - \widehat{v}_h, \boldsymbol{r}\cdot\boldsymbol{n}\rangle_e + \sum_{e\in\Gamma^o}\langle v_h^+ - \widehat{v}_h, \boldsymbol{r}^+\cdot\boldsymbol{n}^+\rangle_{e^+} + \langle v_h^- - \widehat{v}_h, \boldsymbol{r}^-\cdot\boldsymbol{n}^-\rangle_{e^-}\\ &= \langle [\![v_h - \widehat{v}_h]\!], \{\!\{\boldsymbol{r}\}\!\}\rangle_{\Gamma_d} + \sum_{e\in\Gamma^o}\int_{e\in K^+\bigcap K^-}(v_h^+ - \widehat{v}_h)\boldsymbol{r}^+\cdot\boldsymbol{n}^+ + (v_h^- - \widehat{v}_h)\boldsymbol{r}^-\cdot\boldsymbol{n}^- ds.\end{aligned}$$

From the notation of the jump and average we have

$$\boldsymbol{r}^+\cdot\boldsymbol{n}^+ + \boldsymbol{r}^-\cdot\boldsymbol{n}^- = [\![\boldsymbol{r}]\!], \quad \frac{\boldsymbol{r}^+ + \boldsymbol{r}^-}{2} = \{\!\{\boldsymbol{r}\}\!\},$$

then, we solve $\boldsymbol{r}^+\cdot\boldsymbol{n}^+$ and $\boldsymbol{r}^-\cdot\boldsymbol{n}^-$ to get

$$\boldsymbol{r}^+\cdot\boldsymbol{n}^+ = \frac{1}{2}[\![\boldsymbol{r}]\!] + \{\!\{\boldsymbol{r}\}\!\}\cdot\boldsymbol{n}^+, \quad \boldsymbol{r}^-\cdot\boldsymbol{n}^- = \frac{1}{2}[\![\boldsymbol{r}]\!] + \{\!\{\boldsymbol{r}\}\!\}\cdot\boldsymbol{n}^-.$$

Therefore,

$$\begin{aligned}&\int_{e\in K^+\bigcap K^-}(v_h^+ - \widehat{v}_h)\boldsymbol{r}^+\cdot\boldsymbol{n}^+ + (v_h^- - \widehat{v}_h)\boldsymbol{r}^-\cdot\boldsymbol{n}^- ds\\ &= \int_{e\in K^+\bigcap K^-}(v_h^+ - \widehat{v}_h)\left(\frac{1}{2}[\![\boldsymbol{r}]\!] + \{\!\{\boldsymbol{r}\}\!\}\cdot\boldsymbol{n}^+\right) + (v_h^- - \widehat{v}_h)\left(\frac{1}{2}[\![\boldsymbol{r}]\!] + \{\!\{\boldsymbol{r}\}\!\}\cdot\boldsymbol{n}^-\right) ds\\ &= \int_{e\in K^+\bigcap K^-}\{\!\{v_h - \widehat{v}_h\}\!\}[\![\boldsymbol{r}]\!] + [\![v_h - \widehat{v}_h]\!]\cdot\{\!\{\boldsymbol{r}\}\!\} ds.\end{aligned}$$

By the fact that $\Gamma = \Gamma^o\bigcup\Gamma^d$ we finally have

$$\begin{aligned}\langle v_h - \widehat{v}_h, \boldsymbol{r}\cdot\boldsymbol{n}\rangle_{\partial\mathcal{T}_h} &= \langle [\![v_h - \widehat{v}_h]\!], \{\!\{\boldsymbol{r}\}\!\}\rangle_{\Gamma_d} + \langle [\![v_h - \widehat{v}_h]\!], \{\!\{\boldsymbol{r}\}\!\}\rangle_{\Gamma^o} + \langle \{\!\{v_h - \widehat{v}_h\}\!\}, [\![\boldsymbol{r}]\!]\rangle_{\Gamma^o}\\ &= \langle [\![v_h - \widehat{v}_h]\!], \{\!\{\boldsymbol{r}\}\!\}\rangle_{\Gamma} + \langle \{\!\{v_h - \widehat{v}_h\}\!\}, [\![\boldsymbol{r}]\!]\rangle_{\Gamma^o}.\end{aligned}$$

□

2.3 Two lifting operators

In this section we will introduce two lifting operators $R : [L^2(\Gamma)]^d \longrightarrow \boldsymbol{V}_h$ and $L : L^2(\Gamma^o) \longrightarrow \boldsymbol{V}_h$ defined as

$$(R(\boldsymbol{\varphi}), \boldsymbol{r})_\Omega = -\langle \boldsymbol{\varphi}, \{\!\{\boldsymbol{r}\}\!\}\rangle_\Gamma, \quad (L(\psi), \boldsymbol{r})_\Omega = -\langle \psi, [\![\boldsymbol{r}]\!]\rangle_{\Gamma^o}, \quad \forall \boldsymbol{r} \in \boldsymbol{V}_h. \tag{2.7}$$

Add the purpose of the lifting operator (from the view of compactness)

Now we give the bound of these two lifting operators. In what follows we will use the symbols $\lesssim$ to compare quantities which differ only up to positive constants that is independent of the mesh size h or of any function appeared in the estimates.

Lemma 2.3. *Let* $p \in [1, \infty)$ *and* $(v_h, \hat{v}_h) \in W_h \times M_h^g$

$$\|R([\![v_h - \hat{v}_h]\!])\|_{L^p(\Omega)} \lesssim \left(\int_{\Gamma^o} h^{1-p}|[\![v_h]\!]|^p ds\right)^{\frac{1}{p}} + \left(\int_{\Gamma_d} h^{1-p}|v_h - g|^p ds\right)^{\frac{1}{p}},$$

and

$$\|L(\{\!\{v_h-\hat{v}_h\}\!\})\|_{L^p(\Omega)} \lesssim \left(\int_\Gamma h^{1-p}|\{\!\{v_h-\hat{v}_h\}\!\}|^p ds\right)^{\frac{1}{p}}.$$

Especially, for $p=1$

$$\|R([\![v_h-\hat{v}_h]\!])\|_{L^1(\Omega)} \lesssim \int_{\Gamma^o}|[\![v_h]\!]|ds+\int_{\Gamma_d}|v_h-g|ds,$$
$$\|L(\{\!\{v_h-\hat{v}_h\}\!\})\|_{L^1(\Omega)} \lesssim \int_\Gamma|\{\!\{v_h-\hat{v}_h\}\!\}|ds.$$

Proof. We begin by noting that the last two inequalities follow immediately by taking $p=1$ in the first two inequalities. Therefore, it remains to prove the first two results in the Lemma. Following the paper [?], the inf-sup condition of Lemma A2 could be extended to any dimensional space such that there exists constant $C>0$ such that

$$\inf_{\boldsymbol{w}\in\boldsymbol{V}_h}\sup_{\boldsymbol{r}\in\boldsymbol{V}_h}\frac{(\boldsymbol{w},\boldsymbol{r})_\Omega}{\|\boldsymbol{w}\|_{L^p(\Omega)}\|\boldsymbol{r}\|_{L^{p'}(\Omega)}}\geq C>0,$$

where $p\in[1,\infty)$ and the dual index p' satisfies $\frac{1}{p}+\frac{1}{p'}=1$. Fix an arbitrary element $\boldsymbol{w}\in\boldsymbol{V}_h$ this implies

$$\sup_{\boldsymbol{r}\in\boldsymbol{V}_h}\frac{(\boldsymbol{w},\boldsymbol{r})_\Omega}{\|\boldsymbol{w}\|_{L^p(\Omega)}\|\boldsymbol{r}\|_{L^{p'}(\Omega)}}\geq C,$$

i. e. for each $\boldsymbol{w}\in\boldsymbol{V}_h$

$$\|\boldsymbol{w}\|_{L^p(\Omega)}\lesssim\sup_{\boldsymbol{r}\in\boldsymbol{V}_h}\frac{(\boldsymbol{w},\boldsymbol{r})_\Omega}{\|\boldsymbol{r}\|_{L^{p'}(\Omega)}}. \tag{2.8}$$

Taking $\boldsymbol{w}=R([\![v_h-\hat{v}_h]\!])$ we obtain

$$\|R([\![v_h-\hat{v}_h]\!])\|_{L^p(\Omega)}\lesssim\sup_{\boldsymbol{r}\in\boldsymbol{V}_h}\frac{(R([\![v_h-\hat{v}_h]\!]),\boldsymbol{r})_\Omega}{\|\boldsymbol{r}\|_{L^{p'}(\Omega)}}. \tag{2.9}$$

In order to obtain the desired bound we have to find the control of the inner product above

$$\begin{aligned}
&(R([\![v_h-\hat{v}_h]\!]),\boldsymbol{r})_\Omega\\
&=-\langle[\![v_h-\hat{v}_h]\!],\{\!\{\boldsymbol{r}\}\!\}\rangle_\Gamma, && \text{by the definition (2.7)}\\
&=-\langle[\![v_h-\hat{v}_h]\!],\{\!\{\boldsymbol{r}\}\!\}\rangle_{\Gamma^o}-\langle[\![v_h-\hat{v}_h]\!],\{\!\{\boldsymbol{r}\}\!\}\rangle_{\Gamma_d}\\
&=-\langle[\![v_h]\!],\{\!\{\boldsymbol{r}\}\!\}\rangle_{\Gamma^o}-\langle v_h-g_h,\boldsymbol{r}\cdot\boldsymbol{n}\rangle_{\Gamma_d} && \text{by the definition } W_h^g\\
&\leq\int_{\Gamma_o}|[\![v_h]\!]||\{\!\{r\}\!\}|ds+\int_{\Gamma_d}|v_h-g||\boldsymbol{r}|ds\\
&=\int_{\Gamma_o}h^{-\frac{1}{p'}}|[\![v_h]\!]|h^{\frac{1}{p'}}|\{\!\{r\}\!\}|ds+\int_{\Gamma_d}h^{-\frac{1}{p'}}|v_h-g|h^{\frac{1}{p'}}|\boldsymbol{r}|ds\\
&\lesssim\left(\int_{\Gamma_o}h^{-\frac{p}{p'}}|[\![v_h]\!]|^p\right)^{\frac{1}{p}}\left(\int_{\Gamma_o}(h^{\frac{1}{p'}}|\{\!\{r\}\!\}|)^{p'}ds\right)^{\frac{1}{p'}}+\left(\int_{\Gamma_d}h^{-\frac{p}{p'}}|v_h-g|^p\right)^{\frac{1}{p}}\left(\int_{\Gamma_d}(h^{\frac{1}{p'}}|\boldsymbol{r}|)^{p'}ds\right)^{\frac{1}{p'}} && \text{,by Hölder inequality}\\
&\lesssim\left(\left(\int_{\Gamma^o}h^{1-p}|[\![v_h]\!]|^pds\right)^{\frac{1}{p}}+\left(\int_{\Gamma_d}h^{1-p}|v_h-g|^pds\right)^{\frac{1}{p}}\right)\|\boldsymbol{r}\|_{L^{p'}(\Omega)}, && \text{by the inverse property.}
\end{aligned}$$

Inserting the bound into the inequality (2.9) and taking the supremum among all the test functions $\boldsymbol{r}\in\boldsymbol{V}_h$ we prove the first estimate in the Lemma. Similarly, if we take $\boldsymbol{w}=L(\{\!\{v_h-\hat{v}_h\}\!\})$ in (2.8) and follow exactly the same way we will get the bound of the average, which completes the proof. □

2.4 Theorems

In this section, we will summarize the most important theorems that we will use in the proof of the variational convergence. Let us first give some classical results on the L^p- spaces. Check the index of $p \in [1, \infty)$ and give the reference

Theorem 2.4. *Let F_n, $F \in L^p(\Omega)$.*

(1) If

$$F_n \rightharpoonup F \quad \text{weakly in } L^p(\Omega),$$

then

$$\int_\Omega |F|^p \, dx \leq \liminf_{n\to\infty} \int_\Omega |F_n|^p \, dx.$$

(2) If

$$F_n \to F \quad \text{strongly in } L^p(\Omega),$$

then

$$\int_\Omega |F|^p \, dx \to \int_\Omega |F_n|^p \, dx.$$

(3) If $F_n \rightharpoonup F$ weakly in $L^p(\Omega)$ and $F_n \to F$ strongly in $L^p(\Omega)$, then

$$F_n \to F \quad \text{strongly in } L^p(\Omega).$$

Next, we will recall some facts about the space of bounded variation, BV. We first give the definition of the BV space.

Definition 2.1. A function $u \in L^1(\Omega)$ is called a function of bounded variation if there exists a distributional derivative Du of u such that the following equality holds

$$\langle Du, \boldsymbol{\phi} \rangle = -\int_\Omega u \, \nabla\cdot \boldsymbol{\phi} \, dx \qquad \forall \boldsymbol{\phi} \in \boldsymbol{C}_c^1(\Omega).$$

Then, we say that BV space equipped with the norm, $\|u\|_{BV(\Omega)} = \|u\|_{L^1(\Omega)} + |Du|(\Omega)$, is a Banach space. The total variation $|Du|(\Omega)$ is defined by

$$|Du|(\Omega) = \sup_{\substack{\boldsymbol{\phi}\in \boldsymbol{C}_c^1(\Omega) \\ \|\boldsymbol{\phi}\|_{L^\infty(\Omega)} \leq 1}} \int_\Omega u \, \nabla\cdot \boldsymbol{\phi} \, dx,$$

where the boldface symbol $\boldsymbol{C}_c^1(\Omega)$ denotes the vector-valued space of continuously differential functions with compact support in Ω. To achieve the main variational convergence of the minimizing problem the convergence and compactness on BV space will play an important role in our analysis. Let u_n, $u \in BV(\Omega)$.

Definition 2.2. We say that u_n weakly* converges to u in $BV(\Omega)$, denoted by $u_n \overset{*}{\to} u$, if $u_n \to u$ in $L^1_{loc}(\Omega)$ and Du_n weakly converges to Du in the distributional sense.

With the definition given above we have the following compact results

Proposition 2.5. *(1) $u_n \overset{*}{\to} u$ weakly* in BV space if and only if $\{u_n\}$ is bounded in BV space and converges to u in $L^1_{loc}(\Omega)$.*

(2) If $\|u_n\|_{BV(\Omega)} \leq C$ for all $n \geq 1$. Then there is a subsequence u_{n_j} and a function $u \in BV(\Omega)$ such that $u_{n_j} \to u$ in $L^1(\Omega)$.

To conclude this section we give the following Fridrichs inequality for BV space. (See [**?**] P_{839} Lemma 6.)

Lemma 2.6. *Let $u \in BV(\Omega)$ and let Ω_d be a subset of $\partial\Omega$ with positive surface measure. Then there exists a constant C_F such that*

$$\| u \|_{L^1(\Omega)} \leq C_F(|Du|(\Omega) + \| u \|_{L^1(\Omega_d)}), \quad \forall u \in BV(\Omega). \tag{2.10}$$

3 HDG spaces

In this section, we want to put our work in a non-Hilbertian setting which is of independent interest to analyze other nonlinear problems and is more general than that needed to deal with P-Laplacian equation. We only work on the HDG finite element spaces and not in a broken Sobolev spaces although we borrow the techniques from the broken Sobolev spaces, especially the relation between weak derivative and broken derivative by using the lifting operator.(More motivations) For this reason we will equip the HDG finite element space $W_h \times M_h^g$ with the norm

$$\|(v_h, \hat{v}_h)\| := \| \boldsymbol{q}_h \|_{L^p(\Omega)} + (\langle \tau(v_h - \widehat{v}_h), v_h - \widehat{v}_h \rangle_{\partial\mathcal{T}_h})^{\frac{1}{2}}. \tag{3.1}$$

for any $1 \leq p < \infty$. Here the weak representation of the negative gradient, $\boldsymbol{q}_h$, is determined by equation (2.3). With the preliminaries in Section 2 we are now in the position to explicitly express $\boldsymbol{q}_h$ in terms of broken derivative $\nabla_h u_h$ and $\hat{u}_h$. This is very important to start our theoretical analysis.

Lemma 3.1. ***(The characterization of the negative weak gradient $\boldsymbol{q}_h$)***

Suppose $\boldsymbol{q}_h$ is determined by the equation (2.3), then, for every $(v_h, \hat{v}_h) \in W_h \times M_h^g$, $\boldsymbol{q}_h$ can be explicitly expressed as

$$\boldsymbol{q}_h = -(\nabla_h v_h + L(\{\!\{ v_h - \widehat{v}_h \}\!\}) + R([\![v_h - \widehat{v}_h]\!])), \tag{3.2}$$

on each element.

Proof. We start with the definition of $\boldsymbol{q}_h$ in the equation (2.3) and integrate by parts on each element to obtain

$$\begin{aligned}
0 &= (\boldsymbol{q}_h, \boldsymbol{r})_{\mathcal{T}_h} - (v_h, \nabla\cdot \boldsymbol{r})_{\mathcal{T}_h} + \langle \widehat{v}_h, \boldsymbol{r}\cdot\boldsymbol{n} \rangle_{\partial\mathcal{T}_h} \\
&= (\boldsymbol{q}_h, \boldsymbol{r})_{\mathcal{T}_h} - \sum_K (v_h, \nabla\cdot \boldsymbol{r})_K + \sum_K \langle \widehat{v}_h, \boldsymbol{r}\cdot\boldsymbol{n} \rangle_{\partial K} \\
&= (\boldsymbol{q}_h, \boldsymbol{r})_{\mathcal{T}_h} + \sum_K ((\nabla_h v_h, \boldsymbol{r})_K - \langle v_h, \boldsymbol{r}\cdot\boldsymbol{n} \rangle_{dk}) + \sum_K \langle \widehat{v}_h, \boldsymbol{r}\cdot\boldsymbol{n} \rangle_{\partial K} \\
&= (\boldsymbol{q}_h, \boldsymbol{r})_{\mathcal{T}_h} + (\nabla_h v_h, \boldsymbol{r})_{\mathcal{T}_h} - \langle v_h - \widehat{v}_h, \boldsymbol{r}\cdot\boldsymbol{n} \rangle_{\partial\mathcal{T}_h}.
\end{aligned}$$

By Lemma 2.2, then

$$0 = (\boldsymbol{q}_h, \boldsymbol{r})_{\mathcal{T}_h} + (\nabla_h v_h, \boldsymbol{r})_{\mathcal{T}_h} - \langle \{\!\{ v_h - \widehat{v}_h \}\!\}, [\![\boldsymbol{r}]\!] \rangle_{\Gamma^o} - \langle [\![v_h - \widehat{v}_h]\!], \{\!\{ \boldsymbol{r} \}\!\} \rangle_{\Gamma}$$

Recall the definition of lifting operators we have

$$0 = (\boldsymbol{q}_h, \boldsymbol{r})_{\mathcal{T}_h} + (\nabla_h v_h, \boldsymbol{r})_{\mathcal{T}_h} + (L(\{\!\{ v_h - \widehat{v}_h \}\!\}), \boldsymbol{r})_{\mathcal{T}_h} + (R([\![v_h - \widehat{v}_h]\!]), \boldsymbol{r})_{\mathcal{T}_h}.$$

Since $\boldsymbol{r}$ is an arbitrary test function, the following strong elementwise equality

$$\boldsymbol{q}_h = -(\nabla_h v_h + L(\{\!\{ v_h - \widehat{v}_h \}\!\}) + R([\![v_h - \widehat{v}_h]\!])),$$

satisfies on each element. □

To start the analysis of variational convergence it is crucial to lift HDG spaces to be BV spaces, and then use the compactness result on BV spaces to obtain the weakly convergent subsequence, and furthermore we could proceed to get the strongly convergence in the natural function space, say $W^{1,p}(\Omega)$. By this observation we will state our first embedding result. Before we do that we will establish some preliminary bound. First, from the rewritten form of the stability term in Lemma 2.1 we easily see, for each $h \in (0,1]$

$$\langle \tau(v_h - g), v_h - g\rangle_{\Gamma_d} \lesssim \langle \tau(v_h - \widehat{v}_h), u_h - \widehat{v}_h\rangle_{\partial\mathcal{T}_h}, \tag{3.3a}$$
$$\langle \tau[\![\, v_h \,]\!], [\![\, v_h \,]\!]\rangle_{\Gamma^o} \lesssim \langle \tau(v_h - \widehat{v}_h), v_h - \widehat{v}_h\rangle_{\partial\mathcal{T}_h}, \tag{3.3b}$$
$$\langle \tau \{\!\{v_h - \hat{v}_h\}\!\}, \{\!\{v_h - \hat{v}_h\}\!\}\rangle_{\Gamma^o} \lesssim \langle \tau(v_h - \widehat{v}_h), v_h - \widehat{v}_h\rangle_{\partial\mathcal{T}_h}. \tag{3.3c}$$

And because τ is piecewise constant this indicates that

$$\| v_h - g \|_{L^2(\Gamma_d)} \lesssim (\langle \tau(v_h - \widehat{v}_h), v_h - \widehat{v}_h\rangle_{\partial\mathcal{T}_h})^{\frac{1}{2}}, \tag{3.4a}$$
$$\| [\![\, v_h \,]\!] \|_{L^2(\Gamma^o)} \lesssim (\langle \tau(v_h - \widehat{v}_h), v_h - \widehat{v}_h\rangle_{\partial\mathcal{T}_h})^{\frac{1}{2}}, \tag{3.4b}$$
$$\| \{\!\{v_h - \widehat{v}_h\}\!\} \|_{L^2(\Gamma^o)} \lesssim (\langle \tau(v_h - \widehat{v}_h), v_h - \widehat{v}_h\rangle_{\partial\mathcal{T}_h})^{\frac{1}{2}}. \tag{3.4c}$$

Since Ω is bounded domain, we have

$$\| \boldsymbol{q}_h \|_{L^1(\Omega)} \lesssim \| \boldsymbol{q}_h \|_{L^p(\Omega)}, \tag{3.5a}$$
$$\| v_h - g \|_{L^1(\Gamma_d)} \lesssim (\langle \tau(v_h - \widehat{u}_h), v_h - \widehat{v}_h\rangle_{\partial\mathcal{T}_h})^{\frac{1}{2}}, \tag{3.5b}$$
$$\| [\![\, v_h \,]\!] \|_{L^1(\Gamma^o)} \lesssim (\langle \tau(v_h - \widehat{v}_h), v_h - \widehat{v}_h\rangle_{\partial\mathcal{T}_h})^{\frac{1}{2}}, \tag{3.5c}$$
$$\| \{\!\{v_h - \widehat{v}_h\}\!\} \|_{L^1(\Gamma^o)} \lesssim (\langle \tau(v_h - \widehat{v}_h), v_h - \widehat{v}_h\rangle_{\partial\mathcal{T}_h})^{\frac{1}{2}}. \tag{3.5d}$$

Lastly, by Lemma 2.3 we have

$$\|R([\![v_h - \hat{v}_h]\!])\|_{L^1(\Omega)} \lesssim (\langle \tau(v_h - \widehat{v}_h), v_h - \widehat{v}_h\rangle_{\partial\mathcal{T}_h})^{\frac{1}{2}}, \tag{3.6a}$$
$$\|L(\{\!\{v_h - \hat{v}_h\}\!\})\|_{L^1(\Omega)} \lesssim (\langle \tau(v_h - \widehat{u}_h), v_h - \widehat{v}_h\rangle_{\partial\mathcal{T}_h})^{\frac{1}{2}}. \tag{3.6b}$$

Now we are ready to state that if $(v_h, \hat{v}_h)$ belongs to HDG space endowed with the norm defined above in (3.1), then v_h belongs to BV space. In other words HDG space is continuously embedded in BV space.

Theorem 3.2. ***(Embedding Theorem)*** *For each $h \in (0,1]$, let $(v_h, \hat{v}_h) \in W_h \times M_h^g$. There exists a constnat $C > 0$, independent of h such that for all $p \in [1, \infty)$*

$$\| v_h \|_{BV(\Omega)} \leq C \| (v_h, \hat{v}_h) \|. \tag{3.7}$$

Proof. Let us integrate by parts on each element to obtain

$$-\int_\Omega v_h \nabla \cdot \boldsymbol{\varphi} dx = (\nabla_h v_h, \boldsymbol{\varphi})_{\partial\mathcal{T}_h} - \langle [\![\, v_h \,]\!], \boldsymbol{\varphi}\rangle_{\Gamma^o}.$$

Therefore, by Definition 2.1, the total variation is bounded by

$$|Dv_h|(\Omega) \leq \| \nabla_h v_h \|_{L^1(\Omega)} + \| [\![\, v_h \,]\!] \|_{L^1(\Gamma^o)}.$$

By the characterization of $\boldsymbol{q}_h$ in Lemma 3.1, the broken derivative can be rewritten as

$$\nabla_h v_h = -\boldsymbol{q}_h - L(\{\!\{v_h - \widehat{v}_h\}\!\}) - R([\![\, v_h - \widehat{v}_h \,]\!]),$$

on each element. Thus,

$$\begin{aligned}\| \nabla_h v_h \|_{L^1(\Omega)} &\leq \| \boldsymbol{q}_h \|_{L^1(\Omega)} + \|R([\![v_h - \hat{v}_h]\!])\|_{L^1(\Omega)} + \|L(\{\!\{v_h - \hat{v}_h\}\!\})\|_{L^1(\Omega)} \\ &\lesssim \| \boldsymbol{q}_h \|_{L^1(\Omega)} + (\langle \tau(v_h - \widehat{v}_h), v_h - \widehat{v}_h \rangle_{\partial \mathcal{T}_h})^{\frac{1}{2}}, \qquad \text{by (3.6)} \\ &\lesssim \| (v_h, \hat{v}_h) \|.\end{aligned}$$

The last inequality is by the definition of the norm on HDG space. Obviously, the bound of total variation follows immediately by (3.5)

$$|Dv_h|(\Omega) \lesssim \| (v_h, \hat{v}_h) \|.$$

For each $h \in (0, 1]$, by the Fridrichs inequality for BV space in Lemma 2.6

$$\begin{aligned}\| v_h \|_{L^1(\Omega)} &\lesssim |Dv_h|(\Omega) + \| v_h \|_{L^1(\Gamma_d)} \\ &\lesssim |Dv_h|(\Omega) + \| v_h - g \|_{L^1(\Gamma_d)} + \| g \|_{L^1(\Gamma_d)} \\ &\lesssim \| \boldsymbol{q}_h \|_{L^1(\Omega)} + (\langle \tau(v_h - \widehat{v}_h), v_h - \widehat{v}_h \rangle_{\partial \mathcal{T}_h})^{\frac{1}{2}}, \quad \text{by (3.5)}\end{aligned}$$

Once again the definition of the norm on HDG space yields

$$\| v_h \|_{L^1(\Omega)} \lesssim \| (v_h, \hat{v}_h) \|.$$

This completes the proof of boundness. □

(Might more embedding theorem?)

For HDG method, actually we are interested in the sequence of $\{(u_h, \hat{u}_h)\}$ bounded in HDG norm. The second compact result we are going to present below will allow one to extract a weakly* convergent subsequence of $\{u_h\}$ in the BV space and a weakly convergent subsequence of negative weak gradient, $\{\boldsymbol{q}_h\}$, in L^p space. This is actually the compactness result on the HDG space. (rewrite the statement)

Theorem 3.3. ***(Weak compactness)*** *For each $h \in (0, 1]$, let $(u_h, \hat{u}_h \in W_h \times M_h^g$ and $\boldsymbol{q}_h$ be given by (2.3). Suppose the sequence $\{(u_h, \hat{u}_h)\}$ is bounded in HDG norm, i. e. $\sup_h \| (u_h, \hat{u}_h) \| < \infty$ then there exists a subsequence $h_j \downarrow 0$ and a function $u \in W^{1,p}(\Omega)$ with the boundary condition $u = g$ on $\partial\Omega$ such that*

$$\begin{aligned}u_{h_j} &\overset{*}{\rightharpoonup} u \quad \textit{weakly* in } BV(\Omega) \textit{ and} \\ \boldsymbol{q}_{h_j} &\rightharpoonup -\nabla u \quad \textit{weakly in } \boldsymbol{L}^p(\Omega).\end{aligned}$$

Proof. From Theorem 3.2, we know that $\| u_h \|_{BV}$ is bounded. Hence, by Proposition 2.5, there exists a subsequence (which is not relabelled for notational convenience) and a function $u \in BV(\Omega)$ such that $u_h \overset{*}{\rightharpoonup} u$ in BV space, i. e.

$$\begin{aligned}u_h &\to u, \qquad \text{in } L^1(\Omega) \\ Du_h &\rightharpoonup Du, \quad \text{in the distributional sense.}\end{aligned}$$

Since $\sup_h \| (u_h, \hat{u}_h) \| < \infty$, then $\{\boldsymbol{q_h}\}$ is bounded in L^p- space which implies their weak compactness. Upon extracting a further subsequence (again nor relabelled), we obtain that

$$\boldsymbol{q}_h \rightharpoonup q,$$

as $h \to 0$, where $\boldsymbol{q} \in \boldsymbol{L}^p(\Omega)$. We show now that the function $u \in W^{1,p}(\Omega)$ and

$$\boldsymbol{q} = -\nabla u \quad \text{and} \quad u = g \quad \text{on } \partial\Omega$$

We will proceed in two steps. In the first step, we will show that Du_h converges to $-\boldsymbol{q}$ in the distributional sense. Since Du_h converges also to Du in the sense of distributions, it follows that $\boldsymbol{q} = -Du$. Therefore, the singular part of Du is zero, and u has weak derivative $\nabla u = -\boldsymbol{q} \in \boldsymbol{L}^p(\Omega)$. Poincaré's inequality implies that $u \in L^p(\Omega)$ and hence $u \in W^{1,p}(\Omega)$. In the second step, we will prove that u satisfies the boundary condition $u = g$ on $\partial\Omega$.

Step1. Prove that $Du_h \rightharpoonup -\boldsymbol{q}$ in the distributional sense.

For any test function $\boldsymbol{\varphi} \in \boldsymbol{C}_c^1(\Omega)$ and $\boldsymbol{\varphi_h} \in \boldsymbol{V}_h$ we note that

$$
\begin{aligned}
(Du_h, \boldsymbol{\varphi})_\Omega &= -(u_h, \nabla\cdot\boldsymbol{\varphi})_\Omega \\
&= -(u_h, \nabla\cdot(\boldsymbol{\varphi} - \boldsymbol{\varphi_h}))_{\mathcal{T}_h} - (u_h, \nabla\cdot\boldsymbol{\varphi_h})_{\mathcal{T}_h} \\
&= -(u_h, \nabla\cdot(\boldsymbol{\varphi} - \boldsymbol{\varphi_h}))_{\mathcal{T}_h} - (\boldsymbol{q}_h, \boldsymbol{\varphi_h})_{\mathcal{T}_h} - \langle \widehat{u}_h, \boldsymbol{\varphi_h}\cdot\boldsymbol{n}\rangle_{\partial\mathcal{T}_h} && \text{by (2.3)} \\
&= -(\boldsymbol{q}_h, \boldsymbol{\varphi_h})_{\mathcal{T}_h} - (u_h, \nabla\cdot(\boldsymbol{\varphi} - \boldsymbol{\varphi_h}))_{\mathcal{T}_h} + \langle \widehat{u}_h, (\boldsymbol{\varphi} - \boldsymbol{\varphi_h})\cdot\boldsymbol{n}\rangle_{\partial\mathcal{T}_h} - \langle \widehat{u}_h, \boldsymbol{\varphi}\cdot\boldsymbol{n}\rangle_{\partial\mathcal{T}_h} \\
&= -(\boldsymbol{q}_h, \boldsymbol{\varphi_h})_{\mathcal{T}_h} - (u_h, \nabla\cdot(\boldsymbol{\varphi} - \boldsymbol{\varphi_h}))_{\mathcal{T}_h} + \langle \widehat{u}_h, (\boldsymbol{\varphi} - \boldsymbol{\varphi_h})\cdot\boldsymbol{n}\rangle_{\partial\mathcal{T}_h} && \text{by compact support} \\
&= -(\boldsymbol{q}_h, \boldsymbol{\varphi_h})_{\mathcal{T}_h} + (\nabla u_h, \boldsymbol{\varphi} - \boldsymbol{\varphi_h})_{\mathcal{T}_h} - \langle u_h - \widehat{u}_h, (\boldsymbol{\varphi} - \boldsymbol{\varphi_h})\cdot\boldsymbol{n}\rangle_{\partial\mathcal{T}_h} && \text{by integration by parts.}
\end{aligned}
$$

We estimate above three terms separately by taking $\boldsymbol{\varphi_h} = P\varphi$. Then the second term just vanishes. Consider the third term

$$
\begin{aligned}
&|\langle u_h - \widehat{u}_h, (\boldsymbol{\varphi} - \boldsymbol{\varphi_h})\cdot\boldsymbol{n}\rangle_{\partial\mathcal{T}_h}| \lesssim (\langle \tau(u_h - \widehat{u}_h), u_h - \widehat{u}_h\rangle_{\partial\mathcal{T}_h})^{\frac{1}{2}} \| \boldsymbol{\varphi} - \boldsymbol{\varphi_h} \|_{L^2(\partial\mathcal{T}_h)} \quad \text{by the Hölder inequality} \\
&\lesssim h^{\frac{1}{2}} (\langle \tau(u_h - \widehat{u}_h), u_h - \widehat{u}_h\rangle_{\partial\mathcal{T}_h})^{\frac{1}{2}} \|\varphi\|_{H^1(\Omega)}.
\end{aligned}
$$

Therefore, the third term $\langle u_h - \widehat{u}_h, \boldsymbol{\varphi} - \boldsymbol{\varphi_h}\cdot\boldsymbol{n}\rangle_{\partial\mathcal{T}_h}$ converges to 0 as $h \to 0$. By the orthogonality of L^2 projection we have $-(\boldsymbol{q}_h, \boldsymbol{\varphi_h})_{\mathcal{T}_h} = -(\boldsymbol{q}_h, \boldsymbol{\varphi})_{\mathcal{T}_h}$. Since $-(\boldsymbol{q}_h, \boldsymbol{\varphi})_{\mathcal{T}_h}$ converges to $-(\boldsymbol{q}, \boldsymbol{\varphi})_\Omega$ as $h \to 0$ due to the weak convergence of $\boldsymbol{q}_h$ to $\boldsymbol{q}$. Hence, $-(\boldsymbol{q}_h, \boldsymbol{\varphi_h})_{\mathcal{T}_h}$ converges to $-(\boldsymbol{q}, \boldsymbol{\varphi})_\Omega$ as $h \to 0$. Combine the results above we prove that $Du_h \rightharpoonup -\boldsymbol{q}$ in the distributional sense and complete the first part of the proof. Now we continue to prove the Dirichlet boundary condition.

Step2. Prove $u = g$ on $\partial\Omega$.

From the first step we obtain $\boldsymbol{q} = -\nabla u$,then it is not difficult to see that (q, u) also satisfies the weak form (2.3)

$$
(\boldsymbol{q}, \boldsymbol{r})_{\mathcal{T}_h} - (u, \nabla\cdot\boldsymbol{r})_{\mathcal{T}_h} + \langle u, \boldsymbol{r}\cdot\boldsymbol{n}\rangle_{\partial\mathcal{T}_h} = 0, \quad \forall \boldsymbol{r} \in \boldsymbol{V}_h. \tag{3.8}
$$

Subtracting equation (2.3) from equation (3.8) we have the following error equation

$$
(\boldsymbol{q} - \boldsymbol{q}_h, \boldsymbol{r})_{\mathcal{T}_h} - (u - u_h, \nabla\cdot\boldsymbol{r})_{\mathcal{T}_h} + \langle u - \widehat{u}_h, \boldsymbol{r}\cdot\boldsymbol{n}\rangle_{\partial\mathcal{T}_h} = 0, \quad \forall \boldsymbol{r} \in \boldsymbol{V}_h. \tag{3.9}
$$

In order to prove $u = g$ on $\partial\Omega$ it is sufficient to show that

$$
\langle u - g, \boldsymbol{\varphi}\cdot\boldsymbol{n}\rangle_{\Gamma_d} = 0,
$$

for any smooth infinity function $\boldsymbol{\varphi}$. We start by adding and subtracting $\widehat{u}_h$ and proceed as follows

$$
\begin{aligned}
&\langle u - g, \boldsymbol{\varphi}\cdot\boldsymbol{n}\rangle_{\Gamma_d} \\
&= \langle u - \widehat{u}_h, \boldsymbol{\varphi}\cdot\boldsymbol{n}\rangle_{\Gamma_d} + \langle \widehat{u}_h - g, \boldsymbol{\varphi}\cdot\boldsymbol{n}\rangle_{\Gamma_d} \\
&= \langle u - \widehat{u}_h, \boldsymbol{\varphi}\cdot\boldsymbol{n}\rangle_{\partial\mathcal{T}_h} + \langle \widehat{u}_h - g, \boldsymbol{\varphi}\cdot\boldsymbol{n}\rangle_{\Gamma_d} \\
&= \langle \widehat{u}_h - g, \boldsymbol{\varphi}\cdot\boldsymbol{n}\rangle_{\Gamma_d} + \langle u - \widehat{u}_h, (\boldsymbol{\varphi} - \boldsymbol{\varphi}_h)\cdot\boldsymbol{n}\rangle_{\partial\mathcal{T}_h} + \langle u - \widehat{u}_h, \boldsymbol{\varphi_h}\cdot\boldsymbol{n}\rangle_{\partial\mathcal{T}_h} && \text{by adding and subtracting } \boldsymbol{\varphi}_h \\
&= \langle \widehat{u}_h - g, \boldsymbol{\varphi}\cdot\boldsymbol{n}\rangle_{\Gamma_d} + \langle u - \widehat{u}_h, (\boldsymbol{\varphi} - \boldsymbol{\varphi}_h)\cdot\boldsymbol{n}\rangle_{\partial\mathcal{T}_h} \\
&\quad + (u - u_h, \nabla\cdot\boldsymbol{\varphi}_h)_{\mathcal{T}_h} - (\boldsymbol{q} - \boldsymbol{q}_h, \boldsymbol{\varphi}_h)_{\mathcal{T}_h} && \text{by (3.9)} \\
&= \langle \widehat{u}_h - g, \boldsymbol{\varphi}\cdot\boldsymbol{n}\rangle_{\Gamma_d} + \langle u - \widehat{u}_h, (\boldsymbol{\varphi} - \boldsymbol{\varphi}_h)\cdot\boldsymbol{n}\rangle_{\partial\mathcal{T}_h} + (u - u_h, \nabla\cdot(\boldsymbol{\varphi}_h - \boldsymbol{\varphi}))_{\mathcal{T}_h}
\end{aligned}
$$

$$
\begin{aligned}
&+ (u-u_h, \nabla\cdot\boldsymbol{\varphi})_{\mathcal{T}_h} - (\boldsymbol{q}-\boldsymbol{q}_h, \boldsymbol{\varphi}_h-\boldsymbol{\varphi})_{\mathcal{T}_h} - (\boldsymbol{q}-\boldsymbol{q}_h, \boldsymbol{\varphi})_{\mathcal{T}_h} && \text{by adding and subtracting } \boldsymbol{\varphi}_h\\
&= \langle \widehat{u}_h - g, \boldsymbol{\varphi}\cdot\boldsymbol{n}\rangle_{\Gamma_d} + \langle u_h - \widehat{u}_h, (\boldsymbol{\varphi}-\boldsymbol{\varphi}_h)\cdot\boldsymbol{n}\rangle_{\partial\mathcal{T}_h} - (\nabla(u-u_h), \boldsymbol{\varphi}_h-\boldsymbol{\varphi})_{\mathcal{T}_h}\\
&\quad + (u-u_h, \nabla\cdot\boldsymbol{\varphi})_{\mathcal{T}_h} - (\boldsymbol{q}-\boldsymbol{q}_h, \boldsymbol{\varphi}_h-\boldsymbol{\varphi})_{\mathcal{T}_h} - (\boldsymbol{q}-\boldsymbol{q}_h, \boldsymbol{\varphi})_{\mathcal{T}_h} && \text{by integration by parts}\\
&= \langle \widehat{u}_h - g, \boldsymbol{\varphi}\cdot\boldsymbol{n}\rangle_{\Gamma_d} + \langle u_h - \widehat{u}_h, (\boldsymbol{\varphi}-\boldsymbol{\varphi}_h)\cdot\boldsymbol{n}\rangle_{\partial\mathcal{T}_h} - (-\boldsymbol{q}-\nabla u_h, \boldsymbol{\varphi}_h-\boldsymbol{\varphi})_{\mathcal{T}_h} && \text{by } \boldsymbol{q}=-\nabla u\\
&\quad + (u-u_h, \nabla\cdot\boldsymbol{\varphi})_{\mathcal{T}_h} - (\boldsymbol{q}-\boldsymbol{q}_h, \boldsymbol{\varphi}_h-\boldsymbol{\varphi})_{\mathcal{T}_h} - (\boldsymbol{q}-\boldsymbol{q}_h, \boldsymbol{\varphi})_{\mathcal{T}_h}\\
&= \langle \widehat{u}_h - g, \boldsymbol{\varphi}\cdot\boldsymbol{n}\rangle_{\Gamma_d} + \langle u_h - \widehat{u}_h, (\boldsymbol{\varphi}-\boldsymbol{\varphi}_h)\cdot\boldsymbol{n}\rangle_{\partial\mathcal{T}_h} + (\nabla u_h, \boldsymbol{\varphi}_h-\boldsymbol{\varphi})_{\mathcal{T}_h}\\
&\quad + (u-u_h, \nabla\cdot\boldsymbol{\varphi})_{\mathcal{T}_h} + (\boldsymbol{q}_h, \boldsymbol{\varphi}_h-\boldsymbol{\varphi})_{\mathcal{T}_h} - (\boldsymbol{q}-\boldsymbol{q}_h, \boldsymbol{\varphi})_{\mathcal{T}_h}.
\end{aligned}
$$

Suppose we pick $\boldsymbol{\varphi_h}$ be the standard L^2 projection of $\boldsymbol{\varphi}$ then the term $(\nabla u_h, \boldsymbol{\varphi}_h-\boldsymbol{\varphi})_{\mathcal{T}_h}$ and term $(\boldsymbol{q}_h, \boldsymbol{\varphi}_h-\boldsymbol{\varphi})_{\mathcal{T}_h}$ vanish by the orthogonality. $\langle \widehat{u}_h - g, \boldsymbol{\varphi}\cdot\boldsymbol{n}\rangle_{\Gamma_d}$ converges to 0 as $h\to 0$ by the definition of space M_h^g which indicates that $\widehat{u}_h$ converges to g in $L^2(\partial\Omega)$. $(\boldsymbol{q}-\boldsymbol{q}_h, \boldsymbol{\varphi})_{\mathcal{T}_h}$ also converges to 0 by the weak convergence of $\{\boldsymbol{q}_h\}$ in L^p-space. It remains to show $(u-u_h, \nabla\cdot\boldsymbol{\varphi})_{\mathcal{T}_h}$ and $\langle u_h - \widehat{u}_h, (\boldsymbol{\varphi}-\boldsymbol{\varphi}_h)\cdot\boldsymbol{n}\rangle_{\partial\mathcal{T}_h}$. We note that u_h converges to u in $L^1(\Omega)$ space which implies that $(u-u_h, \nabla\cdot\boldsymbol{\varphi})_{\mathcal{T}_h}$ converges to 0 as $h\to 0$. For

$$
|\langle u_h - \widehat{u}_h, (\boldsymbol{\varphi}-\boldsymbol{\varphi}_h)\cdot\boldsymbol{n}\rangle_{\partial\mathcal{T}_h}| \lesssim (\langle \tau(u_h-\widehat{u}_h), u_h-\widehat{u}_h\rangle_{\partial\mathcal{T}_h})^{\frac{1}{2}} h^{k+\frac{1}{2}} \| \varphi \|_{H^{k+1}(\Omega)},
$$

which converges to 0 as $h\to 0$. Finally we prove that

$$
\langle u-g, \boldsymbol{\varphi}\cdot\boldsymbol{n}\rangle_{\Gamma_d} = 0,
$$

for any C^∞ function $\boldsymbol{\varphi}$. This complete the proof. □

4 Variational HDG convergence of P-Laplacian problem

In this section, we will provide a detailed proof of the variational convergence of HDG method for P-Laplacian problem. The method we will apply is so-called direct method. In the calculus of variations, the direct method is a standard tool for constructing a proof of the existence of a minimizer for a given energy functional. The key idea can be summarized as follows:

1. Show that the energy functional is bounded, which allows one to take a minimizing sequence.

2. Show that the minimizing sequence under consideration admits some subsequence that converges to an element on a suitable larger space equipped with a proper topology. The convergence is usually a primary result and in the weak sense.

3. Show that the element obtained in above step actually is the minimizer of the problem and also belongs to the admissible function set.

We will mimic this idea to obtain the variational convergence of discrete energy functional. Let us begin by describing the sketch of our proof. We will first show that the discrete energy function J_h is bounded, and then we will take a minimizing sequence $\{(u_h, \hat{u}_h)\}$. Next, by the compactness established the Section 3 for the HDG space, we will obtain a weakly convergent subsequence $\{(u_{h_j}, \hat{u}_{h_j})\}$ to an element u in $W^{1,p}(\Omega)$ and satisfies the Dirichlet boundary condition. Meanwhile, a related technique, namely $\Gamma-$ convergence, will be applied to prove that u is indeed the solution of the minimizing problem. Moreover, by the dissipative structure of the P-Laplacian problem we actually obtain some extra strong convergence results on the construction of $\boldsymbol{q}_h$ to the negative gradient, $-\nabla u$. And because the exact solution of the P-Laplacian problem is unique, the convergence of the whole sequence of discrete approximation is deduced immediately.

The starting point of our theory is based on the following characterization of the energy functional J_h with the help of trace and lifting operators defined in Section 2. Combining Lemma 3.1 and Lemma 2.1 we have

Lemma 4.1. ***(The second characterization of energy functional*** J_h***)***

The discrete energy functional J_h *can be rewritten as*

$$\begin{aligned} J_h(v_h, \hat{v}_h; \boldsymbol{q}_h) = \frac{1}{p}\int_\Omega |\nabla_h v_h + R([\![v_h - \hat{v}_h]\!]) + L(\{v_h - \hat{v}_h\})|^p \, dx \\ + \langle \tau\{v_h - \widehat{v}_h\}, \{v_h - \widehat{v}_h\}\rangle_{\Gamma^o} + \frac{1}{4}\langle \tau[\![v_h]\!], [\![v_h]\!]\rangle_{\Gamma^o} \\ + \frac{1}{2}\langle \tau(v_h - g), v_h - g\rangle_{\Gamma_d} - \int_\Omega f v_h dx \, dx \end{aligned}$$

4.1 The boundness of discrete energy functional J_h

In order to obtain the boundness of J_h we need the following approximation results.

Lemma 4.2. *Suppose that* $v \in \{C^\infty(\bar{\Omega})$ *and* $v = g$ *on* $\partial\Omega\}$ *then for each* $h \in (0, 1]$ *there exists* $(v_h, \widehat{v}_h) \in W_h \times M_h^g$ *such that*

$$J_h(v_h, \widehat{v}_h; \boldsymbol{q}_h) \to J(v) \quad \text{as } h \to 0. \tag{4.1}$$

Proof. According to the standard polynomial approximation theory(Ciarlet, 1978) there exists $v_h \in W^{1,p}(\Omega) \bigcap W_h$ such that

$$|v - v_h|_{m,p,\Omega} \lesssim h^{k+1-m}|v|_{k+1,p,\Omega}$$

i. e

$$\| v - v_h \|_{W^{1,p}(\Omega)} \xrightarrow{h\to 0} 0,$$

which says that

$$v_h \xrightarrow{h\to 0} v \quad \text{and} \quad \nabla_h v_h \xrightarrow{h\to 0} \nabla v \quad \text{in } L^p \text{ space}$$

Since $v_h \in W^{1,p}(\Omega)$ we know that

$$[\![v_h]\!]|_e = 0, \quad \{\!\{v_h\}\!\}|_e = v_h|_e, \quad \forall e \in \Gamma^o.$$

Therefore, we can choose

$$\widehat{v}_h|_e = v_h|_e, \quad \forall e \in \Gamma^o, \quad \widehat{v}_h|_e = g_h|_e, \quad \forall e \in \Gamma_d. \tag{4.2}$$

Then, in the second form of J_h,

$$\langle \tau\{v_h - \widehat{v}_h\}, \{v_h - \widehat{v}_h\}\rangle_{\Gamma^o} + \frac{1}{4}\langle \tau[\![v_h]\!], [\![v_h]\!]\rangle_{\Gamma^o} = 0.$$

As $v_h \to v$ in L^p space, if $f \in L^{p'}(\Omega)$ then $\int_\Omega f v_h dx$ converges to $\int_\Omega f v dx$ as $h \to 0$. By the choice of the numerical flux in (4.2) and Lemma 2.3 we get

$$L(\{\!\{v_h - \hat{v}_h\}\!\}) = 0, \quad \|R([\![v_h - \hat{v}_h]\!])\|^p_{L^p(\Omega)} \lesssim \int_{\Gamma_d} h^{1-p}|v_h - g|^p ds.$$

For any $p \in [1, \infty)$ if we could show that

$$\|R([\![v_h - \hat{v}_h]\!])\|^p_{L^p(\Omega)} \xrightarrow{h\to 0} 0$$

then, by Lemma 3.1, we obtain

$$\boldsymbol{q}_h \xrightarrow{h\to 0} -\nabla v, \quad \text{in } L^p(\Omega), \tag{4.3}$$

which implies that

$$\frac{1}{p}\int_\Omega |\boldsymbol{q}_h|^p\,dx \xrightarrow{h\to 0} \frac{1}{p}\int_\Omega |\nabla v|^p\,dx.$$

In fact, for each boundary edge $e \in \Gamma_d \bigcap K$, by standard polynomial approximation theory(Ciarlet, 1978) we have estimate

$$\int_e h^{1-p}|v_h - g|^p ds = \int_e h^{1-p}|v_h - v|^p ds \lesssim h|v|^p_{2,p,K} \xrightarrow{h\to 0} 0.$$

Sum over all the boundary edges we obtain that

$$\int_{\Gamma_d} h^{1-p}|v_h - g|^p ds \xrightarrow{h\to 0} 0,$$

which completes the proof of (4.3) and at the same time we also prove fourth term in the second form of J_h converges to 0, i. e

$$\frac{1}{2}\langle \tau(v_h - g), v_h - g\rangle_{\Gamma_d} \xrightarrow{h\to 0} 0.$$

This concludes that

$$J_h(v_h, \widehat{v}_h; \boldsymbol{q}_h) \xrightarrow{h\to 0} J(v).$$

□

By the argument of density we have the following consequence of the above result and the boundness of the discrete energy functional J_h follows right away.

Corollary 4.3. *Suppose that $v \in \{W^{1,p}(\Omega)$ and $v = g$ on $\partial\Omega\}$ for some $p \in [1,\infty)$ then for each $h \in (0,1]$ there exists $(v_h, \widehat{v}_h) \in W_h \times M_h^g$ such that*

$$J_h(v_h, \widehat{v}_h; \boldsymbol{q}_h) \to J(v). \tag{4.4}$$

Proof. Since Ω is assumed to be a Lipschitsz domain, $C^\infty(\bar{\Omega})$ is dense in $W^{1,p}(\Omega)$. And hence we may assume without loss of generality that $v \in C^\infty(\bar{\Omega})$. For such a smooth function, this result follows from standard polynomial approximation theory (Ciarlet, 1978) which is presented in Lemma 4.2. To be complete we sketch the proof as follows. For any $v \in W^{1,p}(\Omega)$ and $v = g$ on $\partial\Omega$ there exists $v_\epsilon \in C^\infty(\bar{\Omega})$ and $v_\epsilon = g$ on $\partial\Omega$ such that

$$\| v - v_\epsilon \|_{W^{1,p}(\Omega)} < \epsilon, \quad \text{and} \quad v = v_\epsilon = g, \quad \text{on} \quad \partial\Omega.$$

This implies that

$$\nabla v_\epsilon \xrightarrow{\epsilon\to 0} \nabla v, \quad v_\epsilon \xrightarrow{\epsilon\to 0} v, \quad \text{in} \quad L^P(\Omega),$$

and

$$J_h(v_\epsilon) \xrightarrow{\epsilon\to 0} J(v).$$

By Lemma 4.2 there exists $(v_h, \widehat{v}_h) \in W_h \times M_h^g$ such that

$$J_h(v_h, \widehat{v}_h; \boldsymbol{q}_h) \xrightarrow{h\to 0} J(v_\epsilon).$$

Thus, we obtain that

$$J_h(v_h, \widehat{v}_h; \boldsymbol{q}_h) \xrightarrow{h\to 0} J(v_\epsilon) \xrightarrow{\epsilon\to 0} J(v),$$

which complete the proof. □

With everything at hand now we can take a minimizing sequence $\{(u_h, \hat{u}_h)\}$ and state the uniform boundness of $J_h(u_h, \widehat{u}_h; \boldsymbol{q}_h)$

Lemma 4.4. *Suppose $(u_h, \widehat{u}_h) \in W_h \times M_h^g$ is the minimizer of the minimizing problem (2.4), then for $h \in (0, 1]$*

$$\sup_h J_h(u_h, \widehat{u}_h; \boldsymbol{q}_h) \leq C, \tag{4.5}$$

where positive constant C is independent of h.

Proof. By Lemma 4.3, for any $v \in \{W^{1,p}(\Omega) \text{ and } v = g \text{ on } \partial\Omega\}$ there exists there exists $(v_h, \widehat{v}_h) \in W_h \times M_h$ such that

$$J_h(v_h, \widehat{v}_h; \boldsymbol{q}_h) \to J(v).$$

Since $(u_h, \widehat{u}_h)$ is the minimizer, then

$$J_h(u_h, \widehat{u}_h; \boldsymbol{q}_h) \leq J_h(v_h, \widehat{v}_h; \boldsymbol{q}_h).$$

Thus, for any $\epsilon > 0$ we obtain that

$$J_h(u_h, \widehat{u}_h; \boldsymbol{q}_h) \leq \epsilon + |J(v)|.$$

Taking $\epsilon \to 0$ we finally obtain

$$\sup_h J_h(u_h, \widehat{u}_h; \boldsymbol{q}_h) \leq |J(v)|.$$

This completes the proof. □

In what follows, we will continue to prove that $\{(u_h, \hat{u}_h)\}$ is uniformly bounded in HDG norm.

4.2 The boundness of the minimizing sequence in HDG norm

We note the expression of the discrete energy function J_h consist of two parts. One is related to the HDG norm we defined in Section 2. That is also how the inspiration comes out. The other one is just the source term $(f, u_h)_\Omega$. Since we already get the uniform boundness of $J_h(u_h, \widehat{u}_h; \boldsymbol{q}_h)$, if we can get the control of the source term $(f, u_h)_\Omega$, then $\{(u_h, \hat{u}_h)\}$ is immediately uniformly bounded in HDG norm. For this purpose we will apply a widely used technique for the Laplacian problem, which is called dual argument. We assume the solution (Φ, Ψ) of the problem

$$\Phi + \nabla\Psi = 0 \quad \text{in } \Omega, \tag{4.6a}$$

$$\nabla \cdot \Phi = f \quad \text{in } \Omega, \tag{4.6b}$$

$$\Psi = 0 \quad \text{on } \partial\Omega, \tag{4.6c}$$

is such that we have the following regularity on a general sobolev space $W^{1,q}(\Omega)$

$$\| \Psi \|_{W^{2,q}(\Omega)} \lesssim \| f \|_{L^q(\Omega)}, \tag{4.7}$$

for any $1 < q < \infty$. (See [?]) Now we can deal with the term $(f, u_h)_\Omega$.

$$\begin{aligned}(f, u_h)_\Omega &= (\nabla\cdot\Phi, u_h)_{\partial\mathcal{T}_h} && \text{by (4.6b)}\\ &= (\nabla\cdot(\Phi - \Phi_h), u_h)_{\partial\mathcal{T}_h} + (\nabla\cdot\Phi_h, u_h)_{\partial\mathcal{T}_h},\end{aligned}$$

$$
\begin{aligned}
&= (\nabla\cdot(\Phi-\Phi_h), u_h)_{\partial\mathcal{T}_h} + (\boldsymbol{q}_h, \Phi_h)_{\mathcal{T}_h} + \langle \widehat{u}_h, \Phi_h\cdot\boldsymbol{n}\rangle_{\partial\mathcal{T}_h} && \text{by (2.3)}\\
&= (\boldsymbol{q}_h, \Phi_h)_{\mathcal{T}_h} + (\nabla\cdot(\Phi-\Phi_h), u_h)_{\partial\mathcal{T}_h} + \langle \widehat{u}_h, (\Phi_h-\Phi)\cdot\boldsymbol{n}\rangle_{\partial\mathcal{T}_h} + \langle \widehat{u}_h, \Phi\cdot\boldsymbol{n}\rangle_{\partial\mathcal{T}_h}\\
&= (\boldsymbol{q}_h, \Phi_h)_{\mathcal{T}_h} + \langle \widehat{u}_h, \Phi\cdot\boldsymbol{n}\rangle_{\partial\mathcal{T}_h} + (\nabla\cdot(\Phi-\Phi_h), u_h)_{\partial\mathcal{T}_h} + \langle \widehat{u}_h, (\Phi_h-\Phi)\cdot\boldsymbol{n}\rangle_{\partial\mathcal{T}_h}\\
&= (\boldsymbol{q}_h, \Phi_h)_{\mathcal{T}_h} + \langle \widehat{u}_h, \Phi\cdot\boldsymbol{n}\rangle_{\partial\mathcal{T}_h} - (\Phi-\Phi_h, \nabla u_h)_{\partial\mathcal{T}_h} + \langle u_h-\widehat{u}_h, (\Phi-\Phi_h)\cdot\boldsymbol{n}\rangle_{\partial\mathcal{T}_h}\\
&= (\boldsymbol{q}_h, \Phi_h)_{\mathcal{T}_h} + \langle g_h, \Phi\cdot\boldsymbol{n}\rangle_{\Gamma_d} - (\Phi-\Phi_h, \nabla u_h)_{\partial\mathcal{T}_h} + \langle u_h-\widehat{u}_h, (\Phi-\Phi_h)\cdot\boldsymbol{n}\rangle_{\partial\mathcal{T}_h}
\end{aligned}
$$

Taking Φ_h as the normal L^2 projection we estimate all the terms above separately.

$$
\begin{aligned}
|(\boldsymbol{q}_h, \Phi_h)_{\mathcal{T}_h}| &= |(\boldsymbol{q}_h, \Phi)_{\mathcal{T}_h} + (\boldsymbol{q}_h, \Phi_h-\Phi)_{\mathcal{T}_h}| && \text{by adding and subtracting}\\
&= |(\boldsymbol{q}_h, \Phi)_{\mathcal{T}_h}| && \text{by orthogonality of } L^2 \text{ projection}\\
&\le \|\, \boldsymbol{q}_h \,\|_{L^p(\Omega)} \|\, \Phi \,\|_{L^{p'}(\Omega)} && \text{by the Hölder inequality}\\
&\le \|\, \boldsymbol{q}_h \,\|_{L^p(\Omega)} \|\, f \,\|_{L^{p'}(\Omega)}, && \text{by regularity of the elliptic problem (4.7).}
\end{aligned}
$$

For the term

$$
\begin{aligned}
&|\langle g_h, \Phi\cdot\boldsymbol{n}\rangle_{\Gamma_d}|\\
&= |\langle g_h-g, \Phi\cdot\boldsymbol{n}\rangle_{\Gamma_d} + \langle g, \Phi\cdot\boldsymbol{n}\rangle_{\Gamma_d}| && \text{by adding and subtracting}\\
&= |\langle g_h-g, \Phi\cdot\boldsymbol{n}\rangle_{\Gamma_d}| + |\langle g, \Phi\cdot\boldsymbol{n}\rangle_{\Gamma_d}|\\
&\le \|\, g-g_h \,\|_{L^2(\partial\Omega)} \|\, \Phi \,\|_{L^2(\partial\Omega)} + \|\, g \,\|_{L^p(\partial\Omega)} \|\, \Phi \,\|_{L^{p'}(\partial\Omega)} && \text{by the Hölder inequality}\\
&\lesssim \|\, g \,\|_{L^2(\partial\Omega)} \|\, \Phi \,\|_{W^{1,2}(\Omega)} + \|\, g \,\|_{L^p(\partial\Omega)} \|\, \Phi \,\|_{W^{1,p'}(\Omega)} && \text{by Trace Theorem}\\
&\lesssim \|\, g \,\|_{L^2(\partial\Omega)} \|\, f \,\|_{L^2(\Omega)} + \|\, g \,\|_{L^p(\partial\Omega)} \|\, f \,\|_{L^{p'}(\Omega)}, && \text{by regularity of the elliptic problem (4.7).}
\end{aligned}
$$

By the definition of L^2 projection the term $(\Phi-\Phi_h, \nabla u_h)_{\partial\mathcal{T}_h} = 0$. For the last term

$$
\begin{aligned}
&|\langle u_h-\widehat{u}_h, (\Phi-\Phi_h)\cdot\boldsymbol{n}\rangle_{\partial\mathcal{T}_h}| \lesssim \|\, \sqrt{\tau}(u_h-\widehat{u}_h) \,\|_{\partial\mathcal{T}_h} \|\, \Phi-\Phi_h \,\|_{\partial\mathcal{T}_h} && \text{by the Hölder inequality}\\
&\lesssim \|\, \sqrt{\tau}(u_h-\widehat{u}_h) \,\|_{\partial\mathcal{T}_h} h^{\frac{1}{2}} \|\, \Phi \,\|_{H^1(\Omega)} && \text{by the standard polynomial approximation theory}\\
&\lesssim \|\, \sqrt{\tau}(u_h-\widehat{u}_h) \,\|_{\partial\mathcal{T}_h} h^{\frac{1}{2}} \|\, f \,\|_{L^2(\Omega)}, && \text{by regularity of the elliptic problem (4.7).}
\end{aligned}
$$

Finally we obtain the following Lemma

Lemma 4.5. ***(Auxiliary control of the source term)***
For any $\epsilon > 0$ there exists a positive constant $C > 0$

$$
|(f, u_h)_\Omega| \le \epsilon \left(\frac{1}{p} \int_\Omega |\boldsymbol{q}_h|^p \, dx + \frac{1}{2} \langle \tau(u_h-\widehat{u}_h), u_h-\widehat{u}_h\rangle_{\partial\mathcal{T}_h} \right) + C,
$$

where C depends only on ϵ, $\|\, g \,\|_{L^p(\partial\Omega)}$, $\|\, f \,\|_{L^{p'}(\Omega)}$, $\|\, g \,\|_{L^2(\partial\Omega)}$ and $\|\, f \,\|_{L^2(\Omega)}$.

Combine Lemma 4.4 and Lemma 4.5 we obtain the boundness of $\{(u_h, \hat{u}_h)\}$ in HDG norm.

Lemma 4.6. ***(The uniform boundness of the minimizing sequence in HDG norm)***
For $h \in (0, 1]$ there exists a positive constant C such that

$$
\sup_h \|\, (u_h, \hat{u}_h) \,\| \le C, \tag{4.8}
$$

where C depends only on $\|\, g \,\|_{L^p(\partial\Omega)}$, $\|\, f \,\|_{L^{p'}(\Omega)}$, $\|\, g \,\|_{L^2(\partial\Omega)}$ and $\|\, f \,\|_{L^2(\Omega)}$.

Finally we come to the point to state the most important variational convergence result in this paper.

4.3 Variational convergence of the discrete minimizing problem

Now we are ready to proof the main Theorem 4.7.

Theorem 4.7. *For each $h \in (0,1]$ and $p \in [1,\infty)$, let $(u_h, \widehat{u}_h) \in W_h \times M_h^g$ be the minimizer of the minimizing problem (2.4). If u is the minimizer of I, then*

$$u_h \xrightarrow{*} u \quad \textit{weakly* in } BV(\Omega) \textit{ and} \tag{4.9a}$$

$$\boldsymbol{q}_h \to -\nabla u \quad \textit{strongly in } \boldsymbol{L}^p(\Omega). \tag{4.9b}$$

$$J_h(u_h, \hat{u}_h; \boldsymbol{q}_h) \to J(u), \tag{4.9c}$$

$$\langle \tau(v_h - \widehat{u}_h), u_h - \widehat{u}_h \rangle_{\partial \mathcal{T}_h} \xrightarrow{h\to 0} 0. \tag{4.9d}$$

Proof. By Corollary 4.3, for each $v \in \{W^{1,p}(\Omega) \text{ and } v = g \text{ on } \partial\Omega\}$ there exists $(v_h, \widehat{v}_h) \in W_h \times M_h^g$ such that

$$J_h(v_h, \widehat{v}_h; \boldsymbol{q}_h) \to J(v).$$

And by Lemma 4.4, $J_h(u_h, \widehat{u}_h; \boldsymbol{q}_h)$ is bounded. Then using in the auxiliary control of the source term $(f, u_h)_{\mathcal{T}_h}$ in Lemma 4.5 we obtain that $(u_h, \widehat{u}_h)$ is bounded in HDG norm, which is Lemma 4.6. By Theorem 3.3, there exists a subsequence u_{h_j} and a function $u \in W^{1,p}(\Omega)$ with boundary condition $u = g$ on $\partial\Omega$ such that

$$u_{h_j} \xrightarrow{*} u \quad \text{weakly* in } BV(\Omega) \text{ and} \tag{4.10}$$

$$\boldsymbol{q}_{h_j} \rightharpoonup -\nabla u \quad \text{weakly in } \boldsymbol{L}^p(\Omega). \tag{4.11}$$

From Section 4.2 it is not difficult to see that

$$(f, u_h)_{\mathcal{T}_h} = (\boldsymbol{q}_h, \Phi)_{\mathcal{T}_h} + \langle g_h, \Phi \cdot \boldsymbol{n} \rangle_{\Gamma_d} + \langle u_h - \widehat{u}_h, (\Phi - \Phi_h) \cdot \boldsymbol{n} \rangle_{\partial \mathcal{T}_h}.$$

By (4.11) we have $(f, u_h)_{\mathcal{T}_h}$ converges to $(-\nabla u, \Phi)_{\mathcal{T}_h} + \langle g, \Phi \cdot \boldsymbol{n} \rangle_{\Gamma_d}$. Integrate by parts we have

$$(-\nabla u, \Phi)_{\mathcal{T}_h} + \langle g, \Phi \cdot \boldsymbol{n} \rangle_{\Gamma_d} = (u, \nabla \cdot \Phi)_{\mathcal{T}_h} - \langle u, \Phi \cdot \boldsymbol{n} \rangle_{\Gamma_d} + \langle g, \Phi \cdot \boldsymbol{n} \rangle_{\Gamma_d} = (f, u),$$

the last equality is due to the dual problem (4.6). Hence, $(f, u_h)_{\mathcal{T}_h}$ converges to (f, u).

To see u is minimizer, fix $v \in \{W^{1,p}(\Omega) \text{ and } v = g \text{ on } \partial\Omega\}$. From Corollary 4.3, $J_h(v_h, \widehat{v}_h; \boldsymbol{q}_h) \to J(v)$, which allows us to estimate

$$\begin{aligned}
J(u) &= \frac{1}{p} \int_\Omega |\nabla u|^p dx - \int_\Omega f u dx \\
&\le \liminf_h \frac{1}{p} \int_\Omega |\boldsymbol{q}_h|^p dx + \liminf_h \int_\Omega -f u_h dx \qquad \text{by Theorem 2.4} \\
&\le \liminf_h \left(\frac{1}{p} \int_\Omega |\boldsymbol{q}_h|^p dx - \int_\Omega f u_h dx \right) \\
&\le \liminf_h J_h(u_h, \hat{u}_h; \boldsymbol{q}_h) \le \limsup_h J_h(u_h, \hat{u}_h; \boldsymbol{q}_h) \\
&\le \limsup_h J_h(v_h, \hat{v}_h; \boldsymbol{q}_h) = J(v).
\end{aligned}$$

Since v is arbitrary we can infer that $J_h(u_h, \hat{u}_h; \boldsymbol{q}_h)$ converges to $J(u)$ by choosing $v = u$. Recall that $(f, u_h)_{\mathcal{T}_h}$ converges to (f, u), then

$$\frac{1}{p} \int_\Omega |\boldsymbol{q}_h|^p \, dx + \frac{1}{2} \langle \tau(v_h - \widehat{v}_h), v_h - \widehat{v}_h \rangle_{\partial \mathcal{T}_h} \to \frac{1}{p} \int_\Omega |\nabla u|^p dx.$$

Note that

$$\begin{aligned}\frac{1}{p}\int_\Omega |\nabla u|^p dx &\le \liminf_h \frac{1}{p}\int_\Omega |\boldsymbol{q}_h|^p dx \\ &\le \limsup_h \frac{1}{p}\int_\Omega |\boldsymbol{q}_h|^p dx \\ &\le \limsup_h \left(\frac{1}{p}\int_\Omega |\boldsymbol{q}_h|^p\, dx + \frac{1}{2}\langle \tau(u_h-\widehat{u}_h), u_h-\widehat{u}_h\rangle_{\partial\mathcal{T}_h}\right) \\ &= \frac{1}{p}\int_\Omega |\nabla u|^p dx,\end{aligned}$$

which implies that

$$\int_\Omega |\boldsymbol{q}_h|^p\, dx \xrightarrow{h\to 0} \int_\Omega |\nabla u|^p dx$$

and

$$\langle \tau(v_h-\widehat{u}_h), u_h-\widehat{u}_h\rangle_{\partial\mathcal{T}_h} \xrightarrow{h\to 0} 0.$$

Therefore, by Theorem 2.4

$$\boldsymbol{q}_h \xrightarrow{h\to 0} -\nabla u, \text{strongly in } \boldsymbol{L}^p(\Omega). \tag{4.12}$$

Since u is the unique minimizer of J, the whole sequence u_h converges to u. □

5 Conclusion

6 Appendix

6.1 Relation between the weak form and discrete energy functional

In this section we will show that the solution the discrete HDG scheme of the P-laplacian is the minimizer of the minimization problem (2.4).

Let $\Phi(\epsilon)$ and $\Psi(\epsilon)$ be functions defined as

$$\Phi(\epsilon) = J_h(u_h+\epsilon w, \hat{u}_h; \boldsymbol{q}_h),\ \Psi(\epsilon) = J_h(u_h, \hat{u}_h+\epsilon\mu; \boldsymbol{q}_h), \tag{6.1}$$

for any $(w,\mu)\in W_h\times M_h^0$ and scalar parameter ϵ. Then $\Phi(\epsilon)$ and $\Psi(\epsilon)$ attain minimum at $\epsilon=0$, i. e

$$\frac{d}{d\epsilon}\Phi(\epsilon)\Big|_{\epsilon=0} = 0,\ \frac{d}{d\epsilon}\Psi(\epsilon)\Big|_{\epsilon=0} = 0. \tag{6.2}$$

We perform the first variation to recover equation (2.1b)-(2.1e) by introducing the intermediate variables σ_h and $\hat{\sigma}$. Denote

$$\delta\boldsymbol{q}_h(u_h,\hat{u}_h;w) = \lim_{\epsilon\to 0}\frac{\boldsymbol{q}_h(u_h+\epsilon w,\hat{u}_h)-\boldsymbol{q}_h(u_h,\hat{u}_h)}{\epsilon}, \tag{6.3}$$

$$\delta\boldsymbol{q}_h(u_h,\hat{u}_h;\mu) = \lim_{\epsilon\to 0}\frac{\boldsymbol{q}_h(u_h,\hat{u}_h+\epsilon\mu)-\boldsymbol{q}_h(u_h,\hat{u}_h)}{\epsilon}, \tag{6.4}$$

for all the test function $(w,\mu)\in W_h\times M_h^o$.

$$0 = \frac{d}{d\epsilon}\Phi(\epsilon)\Big|_{\epsilon=0}$$

$$
\begin{aligned}
&= \frac{1}{p}\int_\Omega \frac{d}{d\epsilon}\left(\left|\boldsymbol{q}_h(u_h+\epsilon w,\hat{u}_h)\right|^p\right)\big|_{\epsilon=0}dx \\
&+ \frac{1}{2}\frac{d}{d\epsilon}\langle\tau(u_h+\epsilon w-\widehat{u}_h),u_h+\epsilon w-\widehat{u}_h\rangle_{\partial\mathcal{T}_h}\big|_{\epsilon=0} - \int_\Omega f\frac{d}{d\epsilon}(u_h+\epsilon w)\big|_{\epsilon=0}dx, \\
&= \left(|\boldsymbol{q}_h|^{p-2}\boldsymbol{q}_h,\delta\boldsymbol{q}_h(u_h,\hat{u}_h;w)\right)_{\mathcal{T}_h} + \langle\tau(u_h-\widehat{u}_h),w\rangle_{\partial\mathcal{T}_h} - (f,w)_{\mathcal{T}_h}.
\end{aligned}
$$

Introduce nonlinear projection σ_h such that

$$
(\boldsymbol{\sigma}_h,\boldsymbol{r})_{\mathcal{T}_h} - (|\boldsymbol{q}_h|^{p-2}\,\boldsymbol{q}_h,\boldsymbol{r})_{\mathcal{T}_h} = 0, \quad \forall\boldsymbol{r}\in\boldsymbol{V}_h. \tag{6.5}
$$

Taking $\boldsymbol{r}=\delta\boldsymbol{q}_h(u_h,\hat{u}_h;w)$ in equation (6.5) we get

$$
\langle|\boldsymbol{q}_h|^{p-2}\boldsymbol{q}_h,\delta\boldsymbol{q}_h(u_h,\hat{u}_h;w)\rangle_{\mathcal{T}_h} = (\boldsymbol{\sigma}_h,\delta\boldsymbol{q}_h(u_h,\hat{u}_h;w))_{\mathcal{T}_h}
$$

From equation (2.3) we have

$$
\begin{aligned}
(\boldsymbol{q}_h(u_h+\epsilon w,\hat{u}_h),\boldsymbol{r})_{\mathcal{T}_h} - (u_h+\epsilon w,\nabla\cdot\boldsymbol{r})_{\mathcal{T}_h} + \langle\widehat{u}_h,\boldsymbol{r}\cdot\boldsymbol{n}\rangle_{\partial\mathcal{T}_h} &= 0, \quad \forall\boldsymbol{r}\in\boldsymbol{V}_h \\
(\boldsymbol{q}_h(u_h,\hat{u}_h),\boldsymbol{r})_{\mathcal{T}_h} - (u_h,\nabla\cdot\boldsymbol{r})_{\mathcal{T}_h} + \langle\widehat{u}_h,\boldsymbol{r}\cdot\boldsymbol{n}\rangle_{\partial\mathcal{T}_h} &= 0, \quad \forall\boldsymbol{r}\in\boldsymbol{V}_h.
\end{aligned}
$$

Hence

$$
0 = \left(\frac{\boldsymbol{q}_h(u_h+\epsilon w,\hat{u}_h)-\boldsymbol{q}_h(u_h,\hat{u}_h)}{\epsilon},\boldsymbol{r}\right)_{\mathcal{T}_h} - \left(\frac{u_h+\epsilon w-u_h}{\epsilon},\nabla\cdot\boldsymbol{r}\right)_{\mathcal{T}_h}, \forall\boldsymbol{r}\in\boldsymbol{V}_h,
$$

i. e

$$
0 = (\delta\boldsymbol{q}_h(u_h,\hat{u}_h;w),\boldsymbol{r})_{\mathcal{T}_h} - (w,\nabla\cdot\boldsymbol{r})_{\mathcal{T}_h}, \quad \forall\boldsymbol{r}\in\boldsymbol{V}_h.
$$

Set $\boldsymbol{r}=\boldsymbol{\sigma}_h$ in above equation

$$
(\boldsymbol{\sigma}_h,\delta\boldsymbol{q}_h(u_h,\hat{u}_h;w))_{\mathcal{T}_h} = (\nabla\cdot\boldsymbol{\sigma_h},w)_{\mathcal{T}_h} = -(\boldsymbol{\sigma}_h,\nabla w)_{\mathcal{T}_h} + \langle\boldsymbol{\sigma_h}\cdot\boldsymbol{n},w\rangle_{\partial\mathcal{T}_h}.
$$

Combine all results we have

$$
\begin{aligned}
0 &= \frac{d}{d\epsilon}\Phi(\epsilon)\Big|_{\epsilon=0} \\
&= \left(|\boldsymbol{q}_h|^{p-2}\boldsymbol{q}_h,\delta\boldsymbol{q}_h(u_h,\hat{u}_h;w)\right)_{\mathcal{T}_h} + \langle\tau(u_h-\widehat{u}_h),w\rangle_{\partial\mathcal{T}_h} - (f,w)_{\mathcal{T}_h} \\
&= (\boldsymbol{\sigma}_h,\delta\boldsymbol{q}_h(u_h,\hat{u}_h;w))_{\mathcal{T}_h} + \langle\tau(u_h-\widehat{u}_h),w\rangle_{\partial\mathcal{T}_h} - (f,w)_{\mathcal{T}_h} \\
&= -(\boldsymbol{\sigma}_h,\nabla w)_{\mathcal{T}_h} + \langle\boldsymbol{\sigma_h}\cdot\boldsymbol{n},w\rangle_{\partial\mathcal{T}_h} + \langle\tau(u_h-\widehat{u}_h),w\rangle_{\partial\mathcal{T}_h} - (f,w)_{\mathcal{T}_h} \\
&= -(\boldsymbol{\sigma}_h,\nabla w)_{\mathcal{T}_h} + \langle\widehat{\boldsymbol{\sigma}}_h\cdot\boldsymbol{n},w\rangle_{\partial\mathcal{T}_h} - (f,w)_{\mathcal{T}_h},
\end{aligned}
$$

where $\widehat{\boldsymbol{\sigma}}_h\cdot\boldsymbol{n}=\boldsymbol{\sigma}_h\cdot\boldsymbol{n}+\tau(u_h-\widehat{u}_h)$. This proves equation (2.1c) in the scheme (2.1). The same technique applied for function $\Psi(\epsilon)$ we could recover the global transmission condition (2.1e) in the scheme (2.1).

$$
\begin{aligned}
0 &= \frac{d}{d\epsilon}\Psi(\epsilon)\Big|_{\epsilon=0} \\
&= \frac{1}{p}\int_\Omega \frac{d}{d\epsilon}\left(\left|\boldsymbol{q}_h(u_h,\hat{u}_h+\epsilon\mu)\right|^p\right)\big|_{\epsilon=0}dx \\
&+ \frac{1}{2}\frac{d}{d\epsilon}\langle\tau(u_h-(\hat{u}_h+\epsilon\mu)),u_h-(\hat{u}_h+\epsilon\mu)\rangle_{\partial\mathcal{T}_h}\big|_{\epsilon=0} - \int_\Omega f\frac{d}{d\epsilon}(u_h)\big|_{\epsilon=0}dx, \\
&= \left(|\boldsymbol{q}_h|^{p-2}\boldsymbol{q}_h,\delta\boldsymbol{q}_h(u_h,\hat{u}_h;\mu)\right)_{\mathcal{T}_h} + \langle\tau(u_h-\widehat{u}_h),-\mu\rangle_{\partial\mathcal{T}_h}.
\end{aligned}
$$

Similarly, we get

$$
(\delta\boldsymbol{q}_h(u_h,\hat{u}_h;\mu),\boldsymbol{r})_{\mathcal{T}_h} + \langle\boldsymbol{r}\cdot\boldsymbol{n},\mu\rangle_{\partial\mathcal{T}_h} = 0, \quad \forall\boldsymbol{r}\in\boldsymbol{V}_h,
$$

from equation (2.3). Set $\boldsymbol{r} = \boldsymbol{\sigma_h}$ to get

$$(\delta \boldsymbol{q}_h(u_h, \hat{u}_h; \mu), \boldsymbol{\sigma_h})_{\mathcal{T}_h} = -\langle \boldsymbol{\sigma_h} \cdot \boldsymbol{n}, \mu \rangle_{\partial \mathcal{T}_h}$$

Recall the definition of variable $\boldsymbol{\sigma_h}$ in the equation (6.5) we get

$$\big(|\boldsymbol{q}_h|^{p-2}\boldsymbol{q}_h, \delta \boldsymbol{q}_h(u_h, \hat{u}_h; \mu)\big)_{\mathcal{T}_h} = (\boldsymbol{\sigma}_h, \delta \boldsymbol{q}_h(u_h, \hat{u}_h; \mu))_{\mathcal{T}_h}.$$

Finally

$$\begin{aligned}
0 &= \frac{d}{d\epsilon}\Psi(\epsilon)\Big|_{\epsilon=0} \\
&= \big(|\boldsymbol{q}_h|^{p-2}\boldsymbol{q}_h, \delta \boldsymbol{q}_h(u_h, \hat{u}_h; \mu)\big)_{\mathcal{T}_h} + \langle \tau(u_h - \widehat{u}_h), -\mu \rangle_{\partial \mathcal{T}_h} \\
&= (\boldsymbol{\sigma}_h, \delta \boldsymbol{q}_h(u_h, \hat{u}_h; \mu))_{\mathcal{T}_h} + \langle \tau(u_h - \widehat{u}_h), -\mu \rangle_{\partial \mathcal{T}_h} \\
&= -\langle \boldsymbol{\sigma_h} \cdot \boldsymbol{n}, \mu \rangle_{\partial \mathcal{T}_h} + \langle \tau(u_h - \widehat{u}_h), -\mu \rangle_{\partial \mathcal{T}_h} \\
&= -\langle \widehat{\boldsymbol{\sigma}}_h \cdot \boldsymbol{n}, \mu \rangle_{\partial \mathcal{T}_h}, \quad \forall \mu \in M_h^o,
\end{aligned}$$

here $\widehat{\boldsymbol{\sigma}}_h \cdot \boldsymbol{n} = \boldsymbol{\sigma}_h \cdot \boldsymbol{n} + \tau(u_h - \widehat{u}_h)$. Finally, we prove that the HDG scheme of the P-Laplacian problem is the Euler Lagrange equation of the discrete energy functional (2.2).